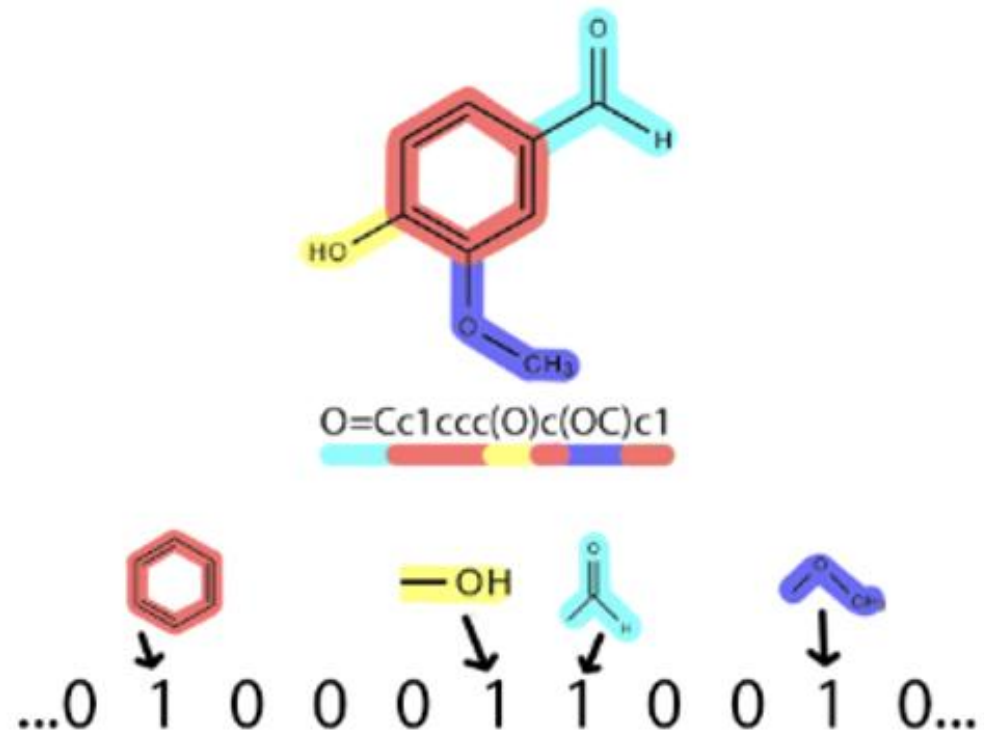


Predicting the activity of a molecule



R software is been used for this data analysis project

- **Introduction:** Machine Learning is widely used in Cheminformatics to predict molecular properties like activity or toxicity. Molecules are often represented by **fingerprints**—binary arrays where each bit indicates the presence of a substructure. **MACCS fingerprints** use 167 predefined bits for this purpose.
- **Aim:** To select the best features that predict the molecular activity against a specific enzyme .
- **Objectives:**
 - To identify the most relevant MACCS fingerprints that effectively predict the molecular activity against the specific enzyme.
 - To explore the relationship between molecular substructures (MACCS fingerprints) and the activity classes (not active, weakly active, strongly active).
 - To visualize the distribution of molecular activity classes and the presence of substructures.
 - To investigate correlations among MACCS fingerprints and their influence on activity labels.
 - To interpret the biological significance of the most predictive substructures in relation to molecular activity.
 - To provide recommendations on how these findings can aid in drug design or chemical synthesis.
- **Methodology:**
 - Data Description
 - Data Cleaning
 - Data Visualization
 - Analytical Techniques
 - Feature Selection
 - Building a model
 - Evaluation
- **Conclusions**

Data Description

Dataset of 2594 molecules - 167 Characteristics - Activity

Column1	zero bit	ISOTOPE	limit the above def'n since
1	0	0	0
2	1	0	0
3	2	0	0
4	3	0	0
5	4	0	0
6	5	0	0
7	6	0	0
8	7	0	0
9	8	0	0
10	9	0	0
11	10	0	0
12	11	0	0
13	12	0	0
14	13	0	0
15	14	0	0

QAA@1	CH3	N	Aromatic	6M Ring	O	Ring	Fragments FIX: this can't be done in SMARTS	Labels
0	1	1	1	1	1	1	0	0.0
0	0	1	1	1	1	1	0	1.0
0	1	1	0	0	1	0	0	0.0
0	1	0	0	1	1	1	0	0.0
0	1	0	0	1	1	1	0	0.0
0	1	0	1	1	1	1	0	1.0
0	1	0	1	1	1	1	0	0.0
0	0	1	0	1	1	1	0	0.0
0	1	0	1	1	1	1	0	1.0
0	0	1	1	1	1	1	0	1.0
0	0	1	1	1	1	1	0	0.0
0	0	1	1	1	1	1	0	0.0
0	0	1	1	1	1	1	0	0.0
0	1	1	1	1	1	1	0	0.0
0	1	1	1	1	1	1	0	1.0

Data Cleaning

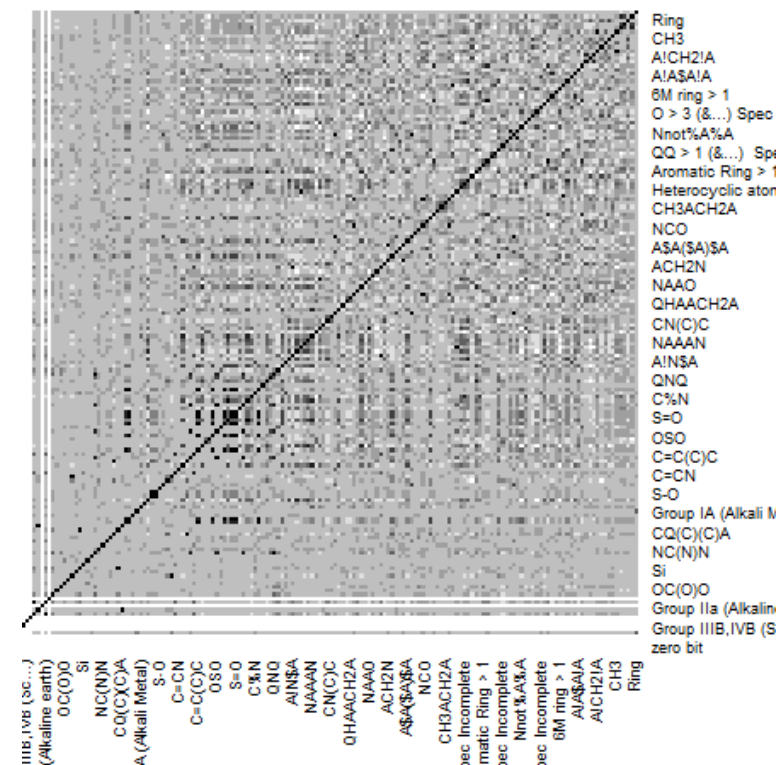
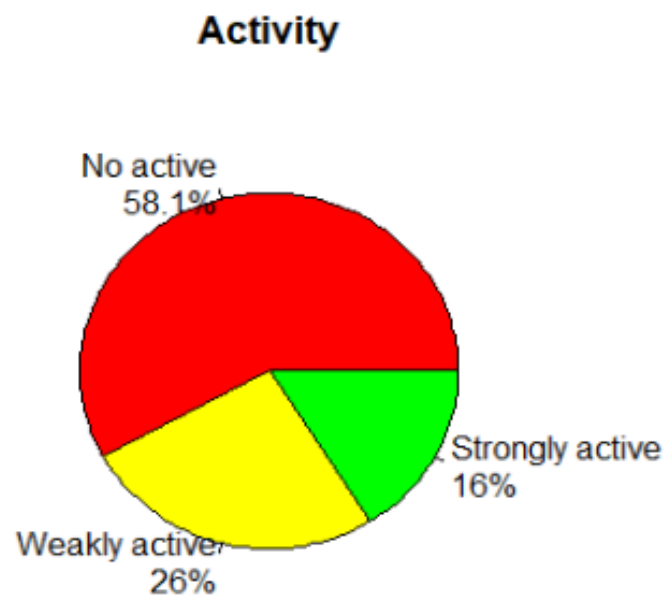
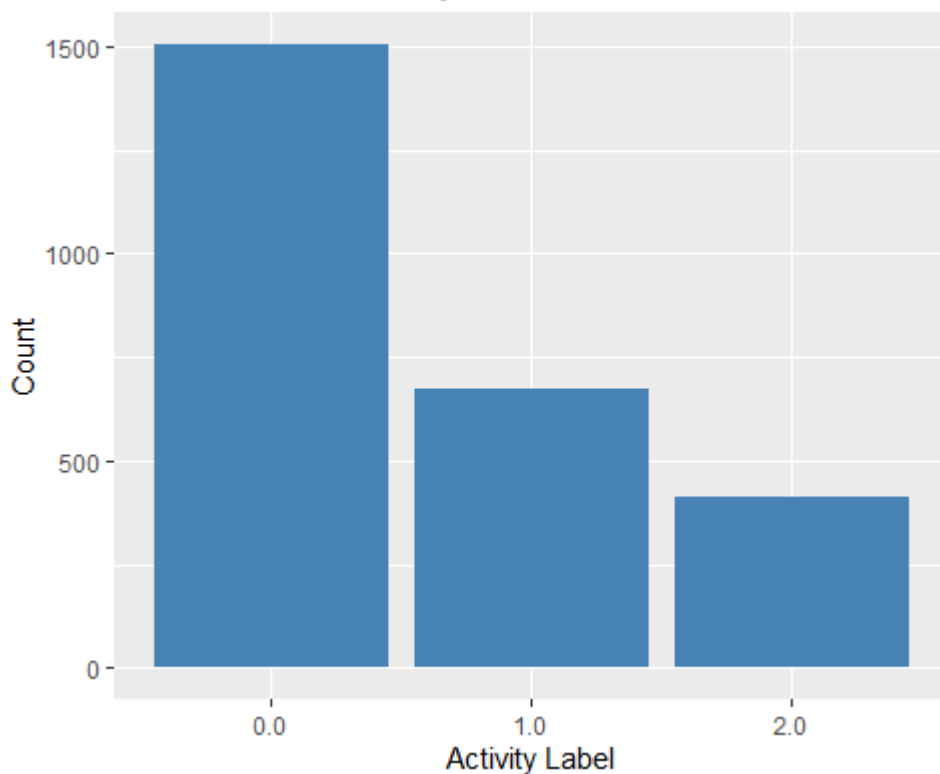
- ✓ Removing the first and last columns
- ✓ Checking for missing values
- ✓ Checking for rows that contain only 0s & 1s
- ✓ Change the name of the columns

▲	feature1 ▾	feature2 ▾	feature3 ▾	feature4 ▾
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0
7	0	0	0	0
8	0	0	0	0
9	0	0	0	0
10	0	0	0	0
11	0	0	0	0
12	0	0	0	0

feature165 ▾	feature166 ▾	feature167 ▾	Labels ▾
1	1	0	0
1	1	0	1
1	0	0	0
1	1	0	0
1	1	0	0
1	1	0	1
1	1	0	0
1	1	0	0
1	1	0	1
1	1	0	1
1	1	0	0
1	1	0	0

Data Visualization

Distribution of Activity Labels



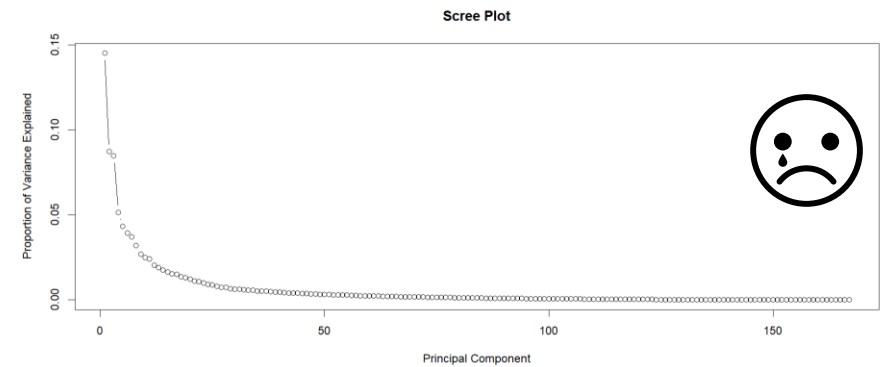
Analytical Techniques

✓ PCA

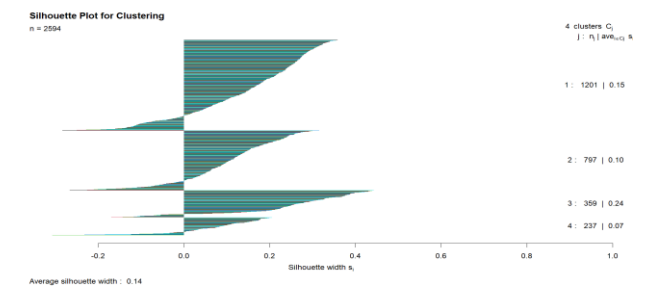
✓ Clustering

- Hierarchical Clustering
- K-Medoids Clustering (PAM) with Jaccard distance
- Spectral Clustering
- DBSC Clustering

✓ Statistical Testing



	Feature	Value
1	P	0.01522771
2	NC(C)N	0.01560922
3	S-O	0.02990153
4	CTN	0.03174464
5	CSO	0.01343907
6	QHAAQH	0.01876037
7	OSO	0.01046368
8	QS	0.04713124
9	CH3ACH3	0.04329196
10	X!A\$A	0.02385198
11	C=C	0.02216533
12	ACH2N	0.04315942
13	CH3AACH2A	0.02401122
14	N=A	0.02310895
15	Aromatic Ring > 1	0.02097800
16	ACH2AACH2A	0.01025105
17	QQ > 1 (&...)	Spec Incomplete 0.03466535
18	QH > 1	0.01273738
19	QH	0.03517818
20	0 > 3 (&...)	Spec Incomplete 0.01635612
21	AQ(A)A	0.02124389
22	QCH2A	0.04452988



Feature Selection

✓ Statistical Testing

✓ Rule mining

✓ Random Forest

✓ Mutual Information

✓ DF of Features

List of rules for better visualization

	NC(C)N=No		QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No	OH=Yes	O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No	QH > 1=Yes	O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes	QS=No	N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes	OSO=No	N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No	CSO=No	QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
P=No	NC(C)N=No		QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes	CH3AACH2A=No	N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No	S-O=No	QHAAQH=Yes		N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0
	NC(C)N=No		QHAAQH=Yes	C=C=No	N=A=No	Aromatic Ring > 1=Yes	QQ > 1 (&.)	Spec Incomplete=No		O > 3 (&.)	Spec Incomplete=No	-->	activity=2.0

Aromatic Ring > 1
QH > 1
OH
QQ > 1 (&...) Spec Incomplete
O > 3 (&...) Spec Incomplete
NC(C)N
QHAAQH
N=A
QS
OSO
CSO
C=C
P
CH3AACH2A
S-O

NC(C)N
QHAAQH
N=A
Aromatic Ring > 1
QQ > 1 (&...) Spec Incomplete
O > 3 (&...) Spec Incomplete



RF Features: "feature124"
 "feature132" "feature63"
 "feature121" "feature76"
 "feature146" "feature151"
 "feature99" "feature80"
 "feature84"

MI Features: "feature34" "feature33"
 "feature68" "feature76" "feature39"
 "feature66" "feature61" "feature62"
 "feature56" "feature59"

"feature39"
 "feature120"
 "feature126"
 "feature131"
 "feature132"
 "feature141"
 "feature55"
 "feature56"
 "feature68"
 "feature76"

Model Building – Data Preparation

Train-Test Split

80%-20% of the dataset

Trainind data 2077 – Testing data 517

Balance Data

58.1% Not Active - 26% Weakly Active - 16% Strongly Active

Overample Strongly Active

Undesample Not Active

691 samples per class

Model Building

- Linear Regression (Baseline)
- Multinomial Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- Neural Network
- Support Vector Machine (SVM)
- Stacking Ensemble Learning

Evaluation metrics

- Accuracy
- Precision
- Recall
- F1-Score (%)
- ROC-AUC

Evaluation

Top 10 Features

Model	Accuracy	Precision	Recall	F1-Score (%)	ROC-AUC
Linear Regression (Baseline)	30.01 %	53.66 %	53.45 %	50.19 %	66.02 %
Multinomial Logistic Regression	51.74 %	49.8 %	24.88 %	21.61 %	47 %
Decision Tree	49.03 %	46.88 %	48.05 %	46.44 %	65.07 %
Random Forest	56.37 %	50.85 %	53.55 %	51.6 %	70.18 %
XGBoost	53.86 %	50.93 %	55.36 %	51.35 %	71.65 %
Neural Network	57.14 %	49.47 %	52.13 %	49.49 %	67.45 %
SVM	56.95 %	51.91 %	53.72 %	52.53 %	70.71 %
Stacking Ensemble Learning	70.68 %	70.63 %	53.93 %	49.61 %	53.93 %

Top 20 Features

Model	Accuracy	Precision	Recall	F1-Score (%)	ROC-AUC
Linear Regression (Baseline)	32.8 %	51.93 %	38.84 %	27.66 %	56.36 %
Multinomial Logistic Regression	51.16 %	49.17 %	53.66 %	49.18 %	66.3 %
Decision Tree	50.19 %	50.47 %	54.78 %	48.16 %	68 %
Random Forest	61.2 %	57.08	62.83 %	57.99 %	76.09 %
XGBoost	58.88 %	55.85 %	61.52 %	56.58 %	74.76 %
Neural Network	52.51 %	53.03 %	57.59 %	51.62 %	73.73 %
SVM	57.72 %	53.44 %	59.09 %	54.31 %	73.2 %
Stacking Ensemble Learning	70.71 %	61.97 %	53.9 %	51.3 %	53.9 %

Improvements

1. Feature Engineering

- Incorporate interaction and polynomial features to better capture complex relationships.
- Perform dynamic feature selection based on feature importance scores.

2. Hyperparameter Tuning

- Optimize model parameters using systematic search strategies within a cross-validation framework.

3. Probabilistic Stacking

- Use predicted class probabilities rather than hard labels in ensemble models to improve performance through increased diversity.

Conclusions

- Preprocessing ensured clean, compatible data; class imbalance was a key challenge
- PCA aided exploration but didn't improve clustering or model performance
- Clustering methods (e.g., DBSCAN, Spectral) failed to produce meaningful groupings
- Feature selection (Chi-square, Apriori, Random Forest, mutual info) revealed key chemical patterns
- Combined sampling strategy (over- and undersampling) addressed class imbalance
- Top-performing models: Random Forest, XGBoost, and SVM**
- Stacking Ensemble had highest accuracy but lower ROC-AUC, indicating weaker minority class detection

Conclusions

Best Performing Models

- Top models: **Random Forest**, **XGBoost**, and **SVM**
- Selected based on accuracy, F1-Score, and ROC-AUC performance
- Provided strong baseline performance but lacked high reliability
- Further improvements needed through **algorithm tuning** and **enhanced feature selection**

Conclusions

Restrictions

- Dataset composed entirely of **binary features**, limiting analytical flexibility
- Techniques like **PCA**, **MCA**, and standard clustering were less effective
- **Low feature variance** reduced usefulness of variance-based elimination
- Binary features lacked **interpretability** compared to continuous data
- **High dimensionality and sparsity** hindered effective visualization and pattern discovery

Conclusions

Difficulties Encountered

- Binary data limited the effectiveness of clustering and dimensionality reduction
- Feature selection methods (Chi-square, Random Forest, Mutual Info) gave inconsistent results
- Class imbalance made it hard to train well-rounded models
- Poor calibration of probabilistic models affected ROC and PR curve interpretation
- Standard accuracy metrics were misleading; required use of balanced and macro-averaged scores
- Computational load increased with larger feature sets, especially in ensemble models
- Stacking ensembles showed potential but added complexity and overfitting risks

This project established a robust, interpretable pipeline for molecular activity classification—highlighting the power of targeted feature selection and rule-based modeling in advancing cheminformatics and laying the groundwork for ILP-integrated discovery.

